Mobile Device Forensics

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Learning Outcomes

- Importance of mobile device forensics
- Diversity of mobile devices
- Fundamental forensic practices for mobile devices
- Smartphones as primary evidence sources
- Technical, legal, and behavioural dimensions

Mobile Device Forensics

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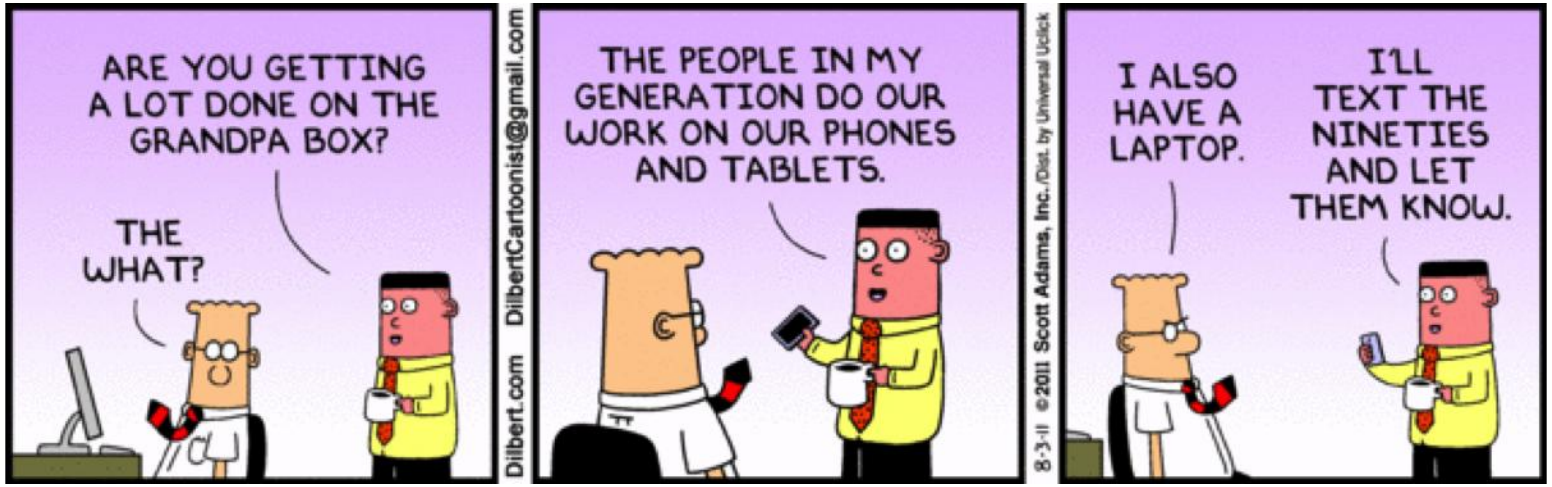
Introduction

In many areas of life and industry, mobile devices are the dominant information processing device…



Communication-Capable Devices ("Phones")

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As of late 2020…

- ≈90% of all criminal investigations in the EU involved mobile devices
- In the UK, this already reached 90% in 2016
- In the same time-frame, US investigations involved mobile devices in ≈75% of cases

While device capabilities of smart phones may rival computers, important characteristics are mobility and (usually) link to cellular network infrastructure

- A large fraction of information relevant to the mobile device is captured by the network (operator) or layered services such as cloud-based systems
- Some of this information may not be immediately obvious, e.g., network operators routinely retaining geolocation traces

Mobile Devices

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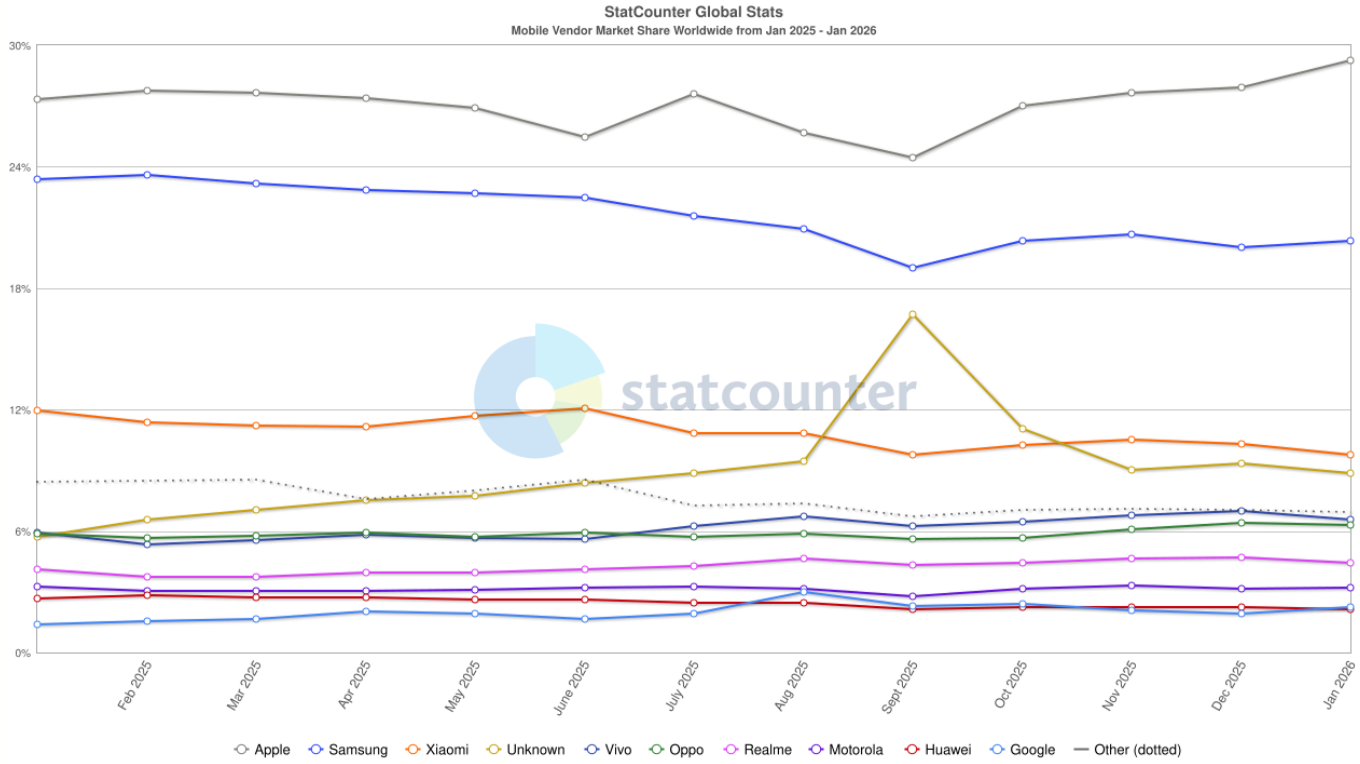
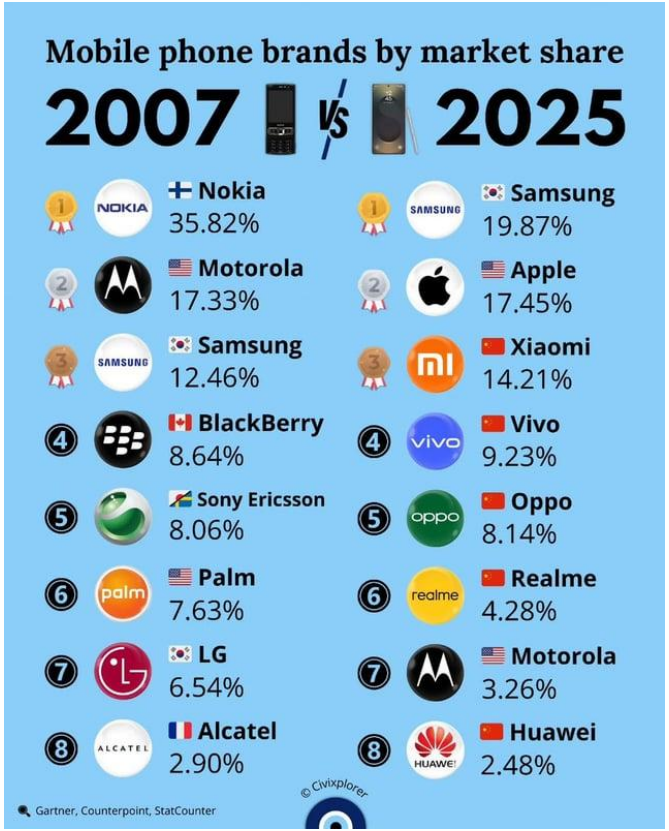
- Variety of shapes and capabilities from simple telephony to general-purpose computing platforms
  - smart phones and tablets
- Non-smart phones have only recently been overtaken by smart phones
- Platforms have effectively been reduced to Apple's iOS and various minor variants of Google Android and Windows/MacOS; others no longer play any meaningful role
- Some differences between devices depending on markets owing to different RF interfaces (2G … 5G, also: CDMA), carrier adaptations
- Compared to operating systems, hardware currently still evolves at significant speed – albeit with diminishing returns

Market Share Evolution (1/3)



Mobile phone brands by market share



Mobile Device Form Factors

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In addition to (largely obsolete) "feature" and smart phones, additional form factors share many if not all characteristics or at least the same operating system:

- **Tablets:** Usually largely indistinguishable in terms of operating system and features
- **Watches:** Relying on mainstream mobile operating systems, stripped-down feature sets and operating systems, except specialized devices (e.g., fitness monitors)
- **Vehicular Systems:** Mobile systems can be used both to operate user interface and entertainment systems in vehicles and also for interfaces (e.g., Android Auto, Apple CarPlay)
- **IoT:** Frequently using Android (if not some custom Linux build or similar)

Different Form Factors

Mobile Device Ecosystem

- Device storage
- SIM card
- SD card
- Application data
- Cloud backups
- Network provider records
- Paired devices - smartwatch, laptop

Types of Mobile Data

User-generated data

- Contacts
- Messages (SMS, WhatsApp, Signal)
- Call logs
- Photos/videos (with EXIF metadata)

System data

- Logs (system events, app usage)
- Device info (IMEI, OS version)
- Network connections

Location data

- GPS traces
- Wi-Fi connections
- Cell tower logs

App data

- Social media (Instagram, Facebook)
- Messaging apps
- Browser history
- Cached files

Mobile Operating Systems

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Android

- Linux-based
- More open ecosystem
- Vendor variability
- Greater file system access
- File system typically EXT4 or F2FS

iOS

- Security-focused design
- Closed ecosystem
- Hardware-backed security
- Encryption by default
- Reliance on backups/cloud

Acquisition Types

Logical Acquisition

- Extract active data via OS APIs
- Faster, limited depth

File System Acquisition

- Access file system structure
- Greater artefact visibility

Physical Acquisition

- Bit-by-bit copy of flash memory
- Increasingly restricted due to encryption

How to extract data

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1. Manual Extraction

Examiner browses device directly

2. Logical Extraction

Uses OS APIs to pull data

3. File System Extraction

Access to file structure

4. Physical Extraction

Bit-by-bit copy of memory

5. Advanced Techniques

JTAG / Chip-off

Encryption in Modern Devices

- Full-disk encryption (Android)
- File-based encryption
- Secure Enclave (iOS)
- Device passcode protection
- Biometric gating

Key Artefact Categories

- Call logs
- SMS/MMS
- Messaging apps (WhatsApp, Signal, Telegram)
- Contacts
- Location data
- Browser history
- App usage logs
- Media metadata
- System logs

Application Forensics

Challenges:

- App encryption
- Cloud synchronisation
- Ephemeral messaging
- Frequent updates

Examples:

- WhatsApp message databases
- Telegram cloud reliance
- Signal encryption constraints

Important areas

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Mobile Metadata

- GPS coordinates
- Device identifiers
- Account IDs
- Timestamps
- App usage patterns

Timeline Reconstruction

- App activity
- System logs
- Network records
- Location services
- Media creation

Challenges in Mobile Forensics

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- Rapid OS updates
- Encryption barriers
- Proprietary ecosystems
- Tool dependency
- Cloud dependency
- Anti-forensics (secure wipe, app deletion)
- Consent vs warrant
- Scope limitations and privacy implications
- Third-party data exposure
- Cross-border data issues

iOS

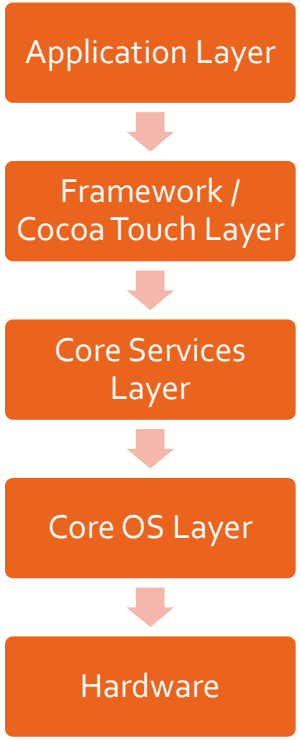
- Derived from OS X
- Interface based on touch and gestures
- In normal operations, iOS uses HFS+ file system
- Can use FAT32 when communicating with a PC

iOS Architecture

- User-installed apps
- Built-in Apple apps
- App sandboxing
- App containers

Forensic relevance

- App data stored in isolated directories
- SQLite databases common
- Plist files
- App-specific encryption



iOS (cont.)

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Cocoa Touch / Framework Layer

User interface frameworks

Touch input handling

Notification services

App lifecycle management

Forensic relevance

- Push notification traces
- Background activity logs
- App state transitions

Core Services Layer

File system access

Database services

Networking stack

Keychain services

Location services

Forensic relevance

- Keychain (credential storage)
- Location artefacts
- System-wide logs
- Cloud synchronisation traces

Core OS Layer

Darwin kernel (XNU)

Memory management

Process scheduling

Security enforcement

Secure Enclave integration

Forensic relevance

- Process control
- Kernel security
- Encryption enforcement
- Boot security chain:

Hardware Layer

Flash storage

Secure Enclave

Biometric sensors

Memory chips

Forensic relevance

- Hardware-backed encryption
- UID keys
- Secure Enclave isolation

Android

- Linux-based operating system, completely open source

Version	Codename	Key Forensic Impact
2.x	Gingerbread	Limited encryption
4.x	KitKat	Improved sandboxing
5.x	Lollipop	**Full-disk encryption introduced**
6.x	Marshmallow	Runtime app permissions
7.x	Nougat	**File-based encryption**
8.x	Oreo	Stronger sandboxing
9	Pie	Encrypted backups
10	—	**Scoped storage introduced**
11	—	**Restricted app access**
12–14	—	**Stronger privacy & security controls**

Full Disk Encryption:
- Entire disk encrypted
- Unlocked once at boot

File Based Encryption:
- Different files unlocked at different times
- Some data accessible before full unlock
- More complex forensic implications

Forensic Implications of Modern Android

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Modern devices often

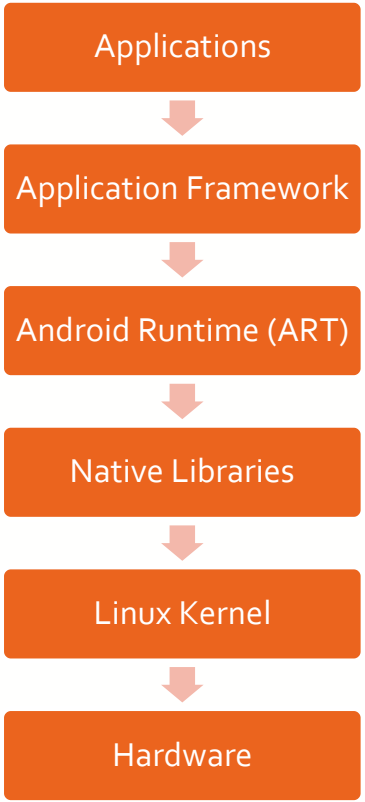
- Require device unlock for full extraction
- Prevent traditional physical acquisition
- Use hardware-backed key storage
- Enforce verified boot

Extraction types available

- Logical
- File system
- Physical

Android Architecture

- Android allows more manufacturer variation
- Core Linux underpinnings remain consistent
- **Forensic relevance**
  - App artefacts stored in /data/data/
  - SQLite databases common
  - Logs in /data/system/
  - Encryption enforced at file system level



Important for Android

- When examining an Android device, always determine:
  - Android version
  - Security patch level
  - Bootloader state
  - Encryption type (FDE vs FBE)
  - Root/jailbreak status

The Process

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Mobile Crime Scene Considerations

- At seizure:
  - Is device powered on?
  - Is it locked?
  - Is it connected to network?
  - Airplane mode?
  - Faraday isolation?

Example:

- iPhone has four-digit pin
  - 10,000 possible combinations of the digits 0–9
- Can use automated process to break iPhone passcode, such as XRY

Rules for Seizing Evidence from a Mobile Device

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- If you plug device into a computer, make sure device does not synchronise with the computer
- Touch evidence as little as possible
- Document what you do to the device
- Don’t accidentally write data to the mobile device

Key Challenges in Mobile Forensics

1. **Encryption**
   i. Full-disk encryption (especially iOS and modern Android)
2. **Rapid OS changes**
   i. Frequent updates break tools and methods
3. **App diversity**
   i. Each app stores data differently
4. **Cloud dependency**
   i. Data may not be on the device at all
5. **Anti-forensics**
   i. Secure deletion
   ii. App-based wiping
   iii. Encrypted messaging (e.g., disappearing messages)

Mobile phone forensic contexts

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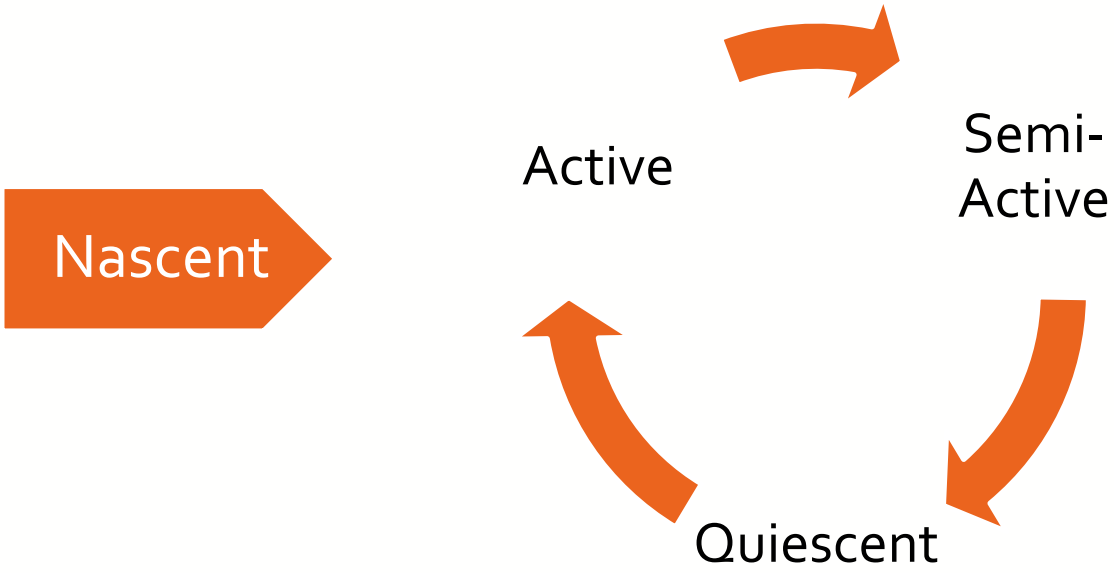
Mobile Device States

Based on NIST Guidelines



Mobile Forensics Tools

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- Mobile forensic tools vary by:
  - Extraction capability (logical / file system / physical)
  - Device support (Android / iOS)
  - Cloud integration
  - Encryption handling
  - Enterprise vs lab use
- No single tool works for all devices.

Popular tools

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Cellebrite (UFED / Physical Analyzer)

Logical, file system, and limited physical extraction

- iOS and Android support
- Cloud artefact parsing
- App-level artefact decoding

Strengths:

- Broad device compatibility
- Regular updates
- Strong reporting tools

Limitations:

- Expensive
- Extraction capability depends on device security level

MSAB (XRY / XAMN)

Logical and file system extraction

- Good device coverage
- Timeline visualisation

Strengths:

- Stable and reliable
- Structured reporting

Limitations:

- Constrained by encryption

Oxygen Forensics Detective

Deep app analysis

- Cloud extraction
- Social media artefact parsing
- Timeline building

Strength:

- Rich artefact visualisation

Limitation:

- Still subject to OS-level encryption restrictions

Open-Source & Research Tools

Andriller

- Android-focused

Autopsy (Mobile modules)

- Limited

ALEAPP / iLEAPP

- Community artefact parsing

The write blocker dilemna

- Use appropriate tools
  - Use controlled APIs or exploits
  - Avoid unnecessary interaction with the device
- Airplane mode / network isolation
  - Prevent remote wiping or syncing
  - Often combined with a Faraday bag
- Avoid unnecessary interaction
  - Don't browse manually unless required
  - Every tap could mean potential data change
- Use read-only approaches where possible
  - Logical acquisition via APIs
  - Bootloader / recovery modes (on some devices)
- Full documentation - some change is inevitable
  - Record device state before acquisition
  - Log every action taken
  - Justify methodology

Discussion Point

Should courts rely on proprietary mobile extraction tools if their internal processes are not fully transparent?

Computer vs Mobile forensics

Aspect	Computer Forensics	Mobile Forensics
Write blocker	Essential hardware	Rarely used
Data state	Static	Dynamic
Risk of change	Low (if blocked)	High (always active OS)
Strategy	Prevent writes	Control + document writes

Case study - WhatsApp Forensics

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What we see?

- Messages are not evidence
- Metadata is not visible
- Context is everything

[10:06, 22/03/2026] Adam: Good morning
[10:06, 22/03/2026] Adam: Did you talk with Charlotte?

[10:06, 22/03/2026] Ellie: Good morning
[10:07, 22/03/2026] Ellie: Yes! She said she wants to go to the cinema

What exists in a forensics context

- Message content
- Sender / receiver identifiers
- Multiple timestamps
- Delivery + read status
- Message IDs
- Linked media

Where Is WhatsApp Data Stored?

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- SQLite databases (chat logs)
- Media folders (images, videos, audio)
- Local backups
- Cloud backups (Google Drive / iCloud)

The layers

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Timestamps

- Sent time
- Delivered time
- Read time
- Device time vs network time

Images

- File creation vs send time
- Metadata (EXIF)
- Thumbnails vs originals

End-to-End Encryption and Biometrics

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- Messages encrypted in transit
- Only sender & receiver can read
- Not readable by network providers
- Decrypted on the device

- WhatsApp can be locked with:
  - Fingerprint
  - Face ID
  - App-level protection (not full device lock)

The takeaway

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- Access depends on:
  - Device state (locked/unlocked)
  - App lock status
  - Backup availability
  - Missed opportunity = lost evidence
- Interpretation is key

Hardware

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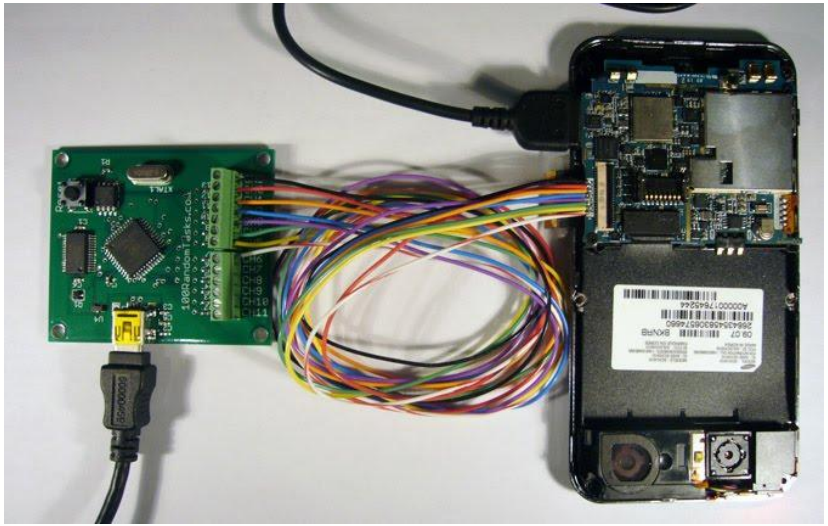
Mobile Device Architecture



- CPU (processing)
- RAM (volatile data)
- Flash storage (persistent data)
- Secure elements (encryption / keys)

Hardware Techniques

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- JTAG (Joint Test Action Group)
  - Hardware-level data extraction
  - Uses test access ports on the device
  - Reads memory directly from the board
  - No need to boot the OS
- Chip-off
  - Physical removal of memory chip
  - Direct reading of flash storage
  - Most invasive technique - High risk of permanent damage
- Damaged devices
  - Physical (water, fire, impact)
  - Electrical damage
  - Corrupted storage
  - Repair to enable logical extraction
  - JTAG / Chip-off
  - Component-level repair
- Locked devices
  - PIN / password
  - Pattern lock
  - Biometrics (fingerprint / face)
  - App-level locks (e.g., WhatsApp)

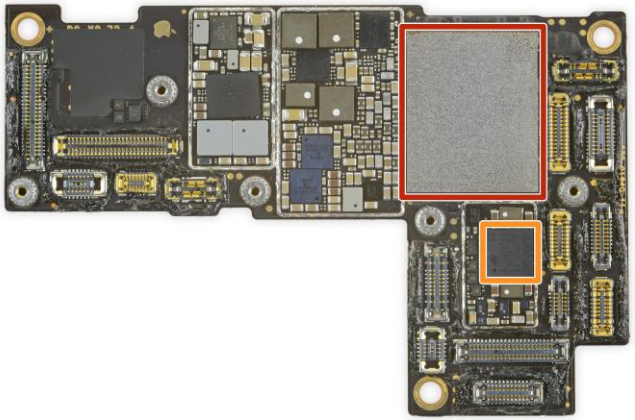
Smart Phone Example 1

iPhone 12 Pro Max with case opened

Smart Phone Example 2

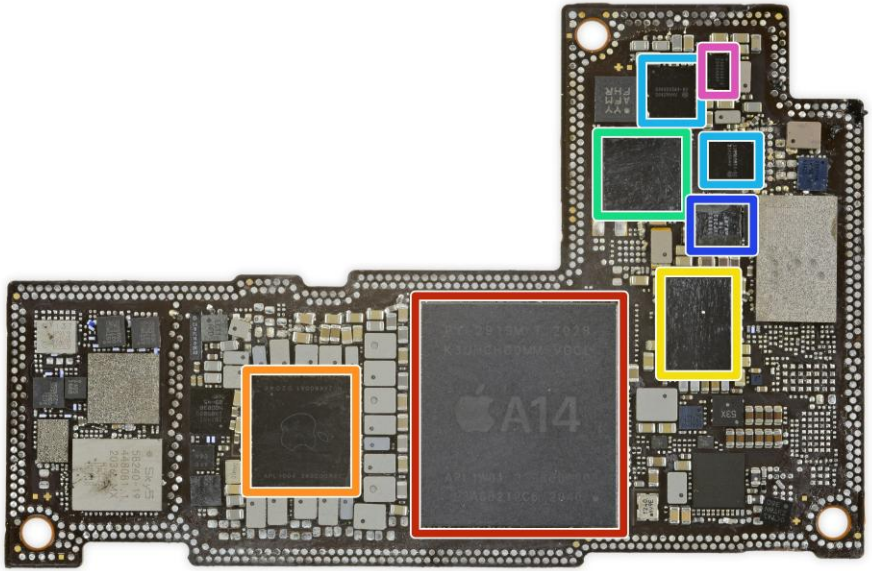
Size of main logic board by comparison

Sliced and separated iPhone 12 Pro Max circuit boards, NAND flash memory marked in red)



Smart Phone Example 3

A14 SoC layered on top of 6GB DRAM (red), power management (orange), wireless power (yellow), audio (green)



SIM Card

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SIM Card Forensics

- Subscriber Identity Module
- Links user to network
- Small storage capacity

Offers:

- IMSI (subscriber identity)
- Phone number
- Contacts (limited)
- SMS (limited)
- Network info

Importance of SIM

- Links suspect to number
- Supports attribution
- Survives device change
- Can corroborate timelines

- Same SIM → multiple phones
- One phone → multiple SIMs
- Impacts attribution

SIM	Device
Subscriber identity	User behaviour
Limited storage	Large storage
Network-related	App + system data
Portable	Device-specific

Summary

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Decision making

Device condition?

Locked or unlocked?

Encryption present?

Risk of data loss?

1. Logical
2. Physical
3. JTAG
4. Chip-off

Summary

- Basic Investigation Techniques
- Low-Level Access Issues
- Unlocking Devices
- Rich but constrained evidence
- OS design shapes outcomes
- Interpretation is critical

Questions?

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Appendix

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Baseline Information - Detailed

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Information may be retained both in the (U)SIM and in the phone's (handset) memory: Device information, call and text message registers for both smart and non-smart phones (but also in cloud accounts!)

- Device identification (MSISDN, IMEI)
- (U)SIM identification (SIM serial, IMSI)
- Recently dialed and received, missed calls (in handset)
- Time and date, duration of these calls or call attempts
- Phone book contacts (SIM, but typically handset)
- Text (SMS) and Multimedia (MMS) messages
- Calendar information
- Service book information
- Roaming and utilization of advanced services

Such data can often be correlated and corroborated by network (operator) data and platform (Google etc.)

Further Mobile Phone Information

Most mobile phones and devices retain further information potentially relevant as evidence:

- Emails
- Web (also WAP) browsing history and cache
- Notes and to-do lists
- Audio clips (particularly voice memos)
- Video clips
- Enhanced telephone book entries, e.g., images
- Documents (e.g., word processor, spreadsheet files)
- Pictures, synchronized and from built-in camera(s)
- Other personal information
- (Instant) messenger application data
- Social network feeds and contacts
- Geolocation (GPS)
- Accelerometer, barometer, etc.
- Service profiles and settings